SUPPORTING ADULT GUIDANCE

Partitioning in more than one way

Place Value · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Exchange Counter Chart, learners see a number as counters in Tens, Ones, Tenths and Hundredths columns. Pressing 1 → 10 breaks one counter into ten in the column to its right; 10 → 1 groups ten back into one. A line under the chart shows the current split (for example 2 + 1.4 + 0.07 = 3.47), and the total never changes. Six target splits are set. Six questions follow, and a stretch explains why an exchange cannot change the number. Learners then find missing parts, judge whether splits really make the number, and invent their own splits under constraints.

Driving Question: *How many ways can I split 3.47?*

The mathematics, in plain terms

Partitioning is not only the standard split by place value (3 + 0.4 + 0.07). Because ten of one column equal one of the column to its left, a number has many partitions, including non-canonical ones such as 2 + 1.4 + 0.07 or 34 tenths + 7 hundredths. This flexibility is the basis of exchanging ("borrowing") in column subtraction and of mental strategies. The chart stops at hundredths; thousandths behave the same way.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Breaking a one

Show that 3.47 = 2 + 1.4 + 0.07.

3.47: counters in each column

Ones		Tenths	Hundredths
3	•	4	7

Break 1 one into 10 tenths

Ones		Tenths	Hundredths
2	•	14	7

The numbers count counters. 2 ones, 14 tenths and 7 hundredths: still 3.47.

1. 3.47 is	3 ones, 4 tenths, 7 hundredths
2. Break 1 one into 10 tenths.	2 ones, 14 tenths, 7 hundredths
3. 14 tenths is	1.4
4. Read the split.	2 + 1.4 + 0.07

3.47 = 2 + 1.4 + 0.07. Taking 1 one away and adding 10 tenths changes nothing.

Example 2 — A missing part

52.6 = 4 tens + ? ones + 6 tenths

52.6

Tens	Ones		Tenths
5	2	•	6

Break 1 ten into 10 ones

Tens	Ones		Tenths
4	12	•	6

The 10 new ones join the 2 already there: 12 ones.

1. Only 4 tens are used, so break the fifth ten.	10 ones
2. Add the ones already there.	10 + 2 = 12
3. Check: 40 + 12 + 0.6	52.6

12 ones.

Exchange first, then explain, then use it

Learners make the exchanges themselves and watch the total stay fixed before any rule is stated. The stretch explains why, using the column relationship from Lesson 4. Missing parts and judging splits then apply the idea; the open splits come last, when learners can check their own.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Exchange Counter Chart and six questions. Stretch: why does an exchange never change the number?	~16 min
Movement	The missing part: eight quick splits.	~7 min
Reflection	Same number? Judge seven splits.	~7 min
Creativity	Split it your way, under constraints.	~10 min
Rest	Many shapes, one amount.	~2 min
Nutrition	Intellectual: flexibility.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a number can only be split one way (3 + 0.4 + 0.07).
Ask: “Could you make 3.47 with only 2 ones? What would have to change?”

Answering 5.8 = 4 + ? with 0.8.
Ask: “Add your parts back up. Do you get 5.8?”

Writing a part in the wrong column (1.08 = 1 + 0.8).
Ask: “Which column is the 8 in? Is 0.8 the same as 8 hundredths?”

Thinking that breaking a counter makes the number bigger, because there are more counters.
Ask: “What is each new counter worth? What are ten of them worth?”

Writing 14 tenths as 0.14.
Ask: “How many tenths make one? So what is 14 tenths?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: tenths and hundredths (Lesson 4). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says 2.63 = 2 + 0.5 + 0.3. The aim is a question that lets them find the error by adding back.

- “Can you add your parts back together?”

- “Which column is the 3 in, in 2.63?”
- “Only 5 tenths are used. Where does the sixth tenth go?”
- “Can you make it on the counter chart?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is this the same as “borrowing” in subtraction?

Yes. Breaking a counter is the exchange used in column subtraction; this lesson builds the idea before it is needed there.

Why does the chart stop at hundredths?

To keep the counters countable. The same exchange works between hundredths and thousandths.

Is the stretch required?

No. The secure core is making and checking non-standard splits and finding missing parts.

What comes next?

Lesson 6: which is larger? Comparing and ordering numbers, including decimals.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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